THE DUAL OF A PERFECT COMPLEX IS PERFECT AND DOUBLE DUALIZATION IS THE IDENTITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Evaluation map. Cohomology of Sheaves, Lemma cohomology-lemma-internal-hom-evaluate. The canonical map into the double dual.
- Perfect case of the evaluation map. Cohomology of Sheaves, Lemma cohomology-lemma-internal-hom-evaluate-isom. The map is an isomorphism for perfect K.
- Internal hom from strictly perfect. Cohomology of Sheaves, Lemma cohomology-lemma-Rhom-strictly-perfect. K$\widehat{\ }$ or represented by termwise duals.
- Evaluation map from the dual. Cohomology of Sheaves, Lemma cohomology-lemma-dual. The map (equation-eval).

Source reference. Original lemma 0A8U: sources/stacks-chow/tags.txt, source lines 12927–12927. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0G40: sources/stacks-chow/tags.txt, source lines 20110–20110. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DL: sources/stacks-chow/tags.txt, source lines 10641–10641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08I1: sources/stacks-chow/tags.txt, source lines 10799–10799. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 08I2: sources/stacks-chow/tags.txt, source lines 10800–10800. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let K be a perfect object of $D(\mathcal{O}_X)$. Then $K^\vee = R\mathcal{H}om(K, \mathcal{O}_X)$ is a perfect object too and $(K^\vee)^\vee \cong K$. There are functorial isomorphisms*

$$M \otimes^{\mathbf{L}}_{\mathcal{O}_X} K^\vee = R\mathcal{H}om(K, M)$$

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and

$$H^0(X, M \otimes^{\mathbf{L}}_{\mathcal{O}_X} K^\vee) = \mathrm{Hom}_{D(\mathcal{O}_X)}(K, M)$$

for M in $D(\mathcal{O}_X)$.

Proof. By Lemma 0A8U there is a canonical map

$$K = R\mathcal{H}om(\mathcal{O}_X, \mathcal{O}_X) \otimes^{\mathbf{L}}_{\mathcal{O}_X} K \longrightarrow R\mathcal{H}om(R\mathcal{H}om(K, \mathcal{O}_X), \mathcal{O}_X) = (K^\vee)^\vee$$

which is an isomorphism by Lemma 0G40. To check the other statements we will use without further mention that formation of internal hom commutes with restriction to opens (Lemma 08DL). We may check $K^\vee$ is perfect locally on X. By Lemma 08I1 to see the final statement it suffices to check that the map (08I2)

$$M \otimes^{\mathbf{L}}_{\mathcal{O}_X} K^\vee \longrightarrow R\mathcal{H}om(K, M)$$

is an isomorphism. This is local on X as well. Hence it suffices to prove these two statements K is represented by a strictly perfect complex.

Assume K is represented by the strictly perfect complex $\mathcal{E}^\bullet$. Then it follows from Lemma 08DM that $K^\vee$ is represented by the complex whose terms are $(\mathcal{E}^{-n})^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^{-n}, \mathcal{O}_X)$ in degree n. Since $\mathcal{E}^{-n}$ is a direct summand of a finite free $\mathcal{O}_X$-module, so is $(\mathcal{E}^{-n})^\vee$. Hence $K^\vee$ is represented by a strictly perfect complex too and we see that $K^\vee$ is perfect. To see that (08I2) is an isomorphism, represent M by a complex $\mathcal{F}^\bullet$. By Lemma 08DM the complex $R\mathcal{H}om(K, M)$ is represented by the complex with terms

$$\bigoplus\nolimits_{n=p+q} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}^{-q}, \mathcal{F}^p)$$

On the other hand, the object $M \otimes^{\mathbf{L}}_{\mathcal{O}_X} K^\vee$ is represented by the complex with terms

$$\bigoplus\nolimits_{n=p+q} \mathcal{F}^p \otimes_{\mathcal{O}_X} (\mathcal{E}^{-q})^\vee$$

Thus the assertion that (08I2) is an isomorphism reduces to the assertion that the canonical map

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X) \longrightarrow \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{F})$$

is an isomorphism when $\mathcal{E}$ is a direct summand of a finite free $\mathcal{O}_X$-module and $\mathcal{F}$ is any $\mathcal{O}_X$-module. This follows immediately from the corresponding statement when $\mathcal{E}$ is finite free. □